

Pod Name	Node	GPU	Startup (sec)
qwen-qwe-c9c8568d-en3-0-6b-decode-7bbc774556-gr4zl	pokprod-b93r38s3	NVIDIA-H100-80GB-HBM3	144.0s
qwen-qwe-c9c8568d-en3-0-6b-decode-7bbc774556-z4gt5	pokprod-b93r38s3	NVIDIA-H100-80GB-HBM3	145.0s
qwen-qwe-c9c8568d-en3-0-6b-decode-7bbc774556-tvjs4	pokprod-b93r43s0	NVIDIA-H100-80GB-HBM3	5037.0s

EPP Configuration (Feature Gates & Scorer Weights)

apiVersion: inference.networking.x-k8s.io/v1alpha1
kind: EndpointPickerConfig
featureGates:
- flowControl
plugins:
- type: queue-scorer
- type: kv-cache-utilization-scorer
- type: prefix-cache-scorer
schedulingProfiles:
- name: default
 plugins:
 - pluginRef: queue-scorer
 weight: 2
 - pluginRef: kv-cache-utilization-scorer
 weight: 2
 - pluginRef: prefix-cache-scorer
 weight: 3

Benchmark Load Generator Configuration

Profile: chatbot_synthetic.yaml
target: http://infra-llmdbench-inference-gateway-istio.wva-bench-test.svc.cluster.local:80
model: Qwen/Qwen3-0.6B
request_type: text_completions
profile: constant
rate: [1,2,4,8]
max_seconds: 600
random_seed: 42
data:
 prompt_tokens_min: 10
 prompt_tokens_max: 8192
 prompt_tokens: 4096
 prompt_tokens_stdev: 2048
 output_tokens_min: 10
 output_tokens_max: 2048
 output_tokens: 1024
 output_tokens_stdev: 512
 samples: 1000

Autoscaling Configuration (HPA & VA)

Variant (VA) [workload-variant-autoscaler-va]: Min Replicas = 1, Max Replicas = 2, Cost Factor = 10.0
HorizontalPodAutoscaler (HPA) [workload-variant-autoscaler-hpa]: Min Replicas = 1, Max Replicas = 10
 > ScaleUp: Stabilization Window = 240s | Policies = [{"periodSeconds": 150, "type": "Pods", "value": 10}]
 > ScaleDown: Stabilization Window = 240s | Policies = [{"periodSeconds": 150, "type": "Pods", "value": 10}]

True Serving Capacity Analysis (GuideLLM results.json)

Rate: 1.0 RPS	Achieved: 0.99 RPS	Errors: 0.00 RPS	Tokens/s: 5146.79
Rate: 2.0 RPS	Achieved: 1.98 RPS	Errors: 0.00 RPS	Tokens/s: 10124.93
Rate: 4.0 RPS	Achieved: 3.91 RPS	Errors: 0.04 RPS	Tokens/s: 20225.93
Rate: 8.0 RPS	Achieved: 5.41 RPS	Errors: 2.24 RPS	Tokens/s: 27786.18

=> Estimated True Serving Capacity: ~1.98 RPS (Highest successful rate with 0 errors)

Understanding GuideLLM Metrics:

- Target Rate (RPS): The configured constant request rate the load generator attempts to send.
- Achieved RPS: The actual rate of requests that completed successfully with a full response.
- Error RPS: The rate of requests that failed (e.g., HTTP 5xx) or were strictly dropped/incomplete.
- True Serving Capacity: Evaluated as the highest Achieved RPS recorded just before the system reaches hardware saturation and begins generating Error RPS (dropped requests).

Autoscaling Run Score (Lower is Better)

Worst-Case P99 TTFT: 67948.06 ms
Worst-Case P99 ITL: 31.41 ms
Average Replicas: 1.22

Target SLAs: TTFT = 50.0ms | ITL = 50.0ms

Formula Engine:

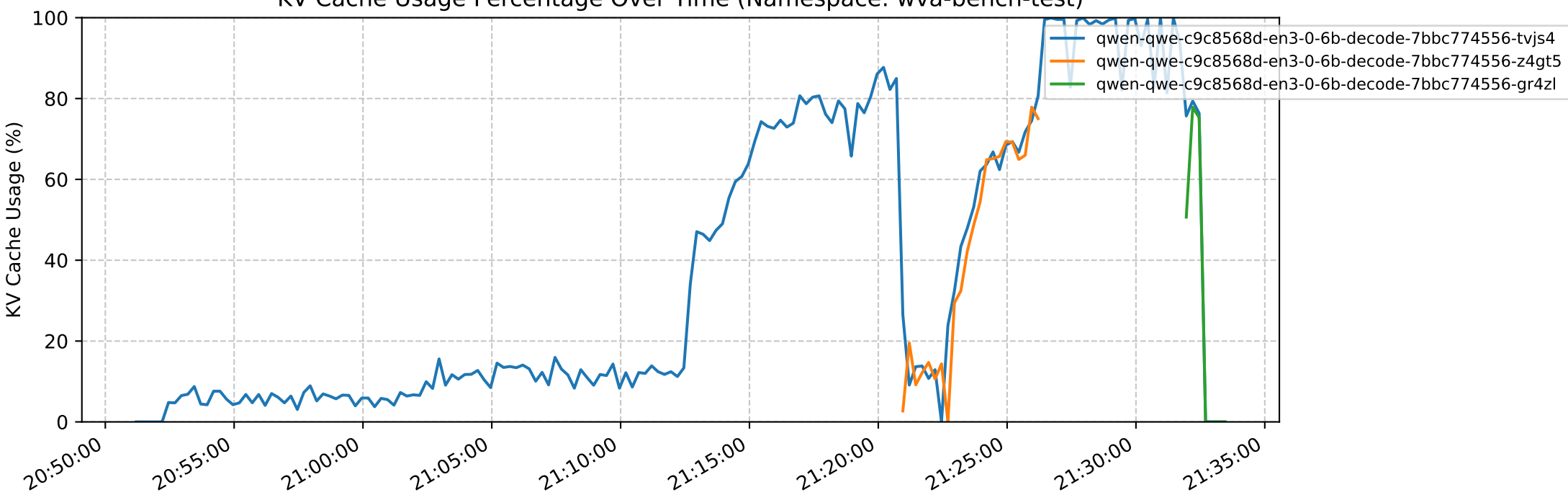
1. TTFT Penalty = (Actual P99 TTFT) / (SLA TTFT)
 2. ITL Penalty = (Actual P99 ITL) / (SLA ITL)
 3. Cost Factor = VA Custom Resource declared Cost (10.0)
 4. Avg Replicas Factor = Average running decode pods (1.22)
- Autoscaling Score = Avg_Replicas × Cost_Factor × (TTFT_Penalty + ITL_Penalty)

Latency Penalty Subtotal = (67948.06/50.0) + (31.41/50.0) = 1359.59
Resource Multiplier = 1.22 × 10.0 = 12.24
=> Final Score = 12.24 × 1359.59 = 16634.97

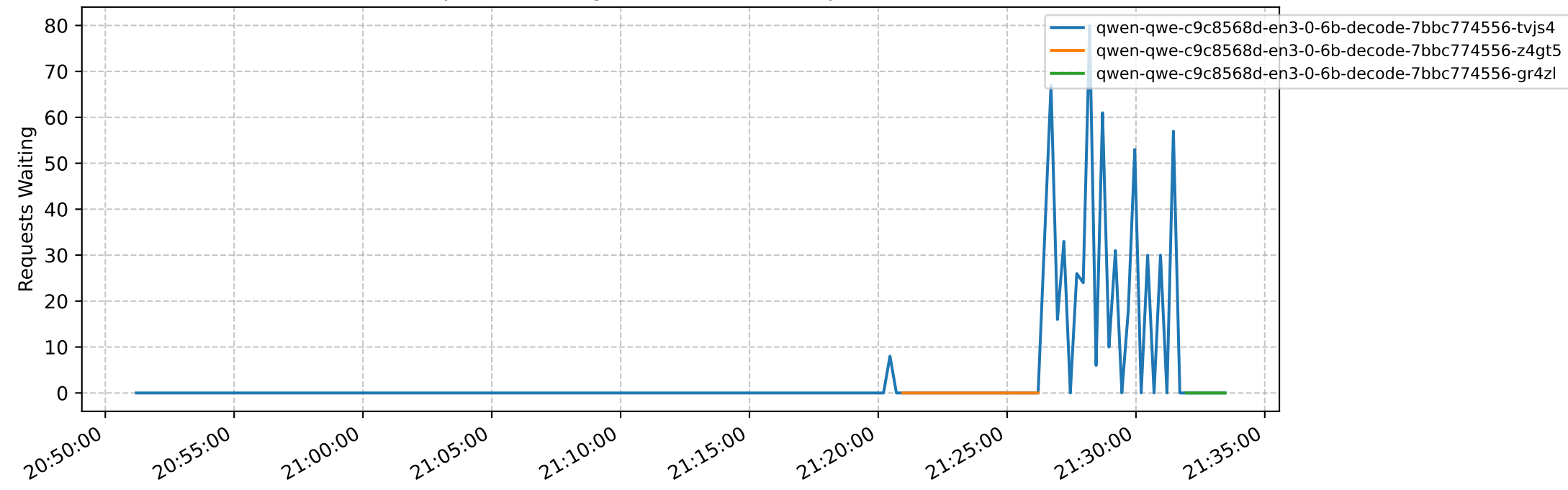
GuideLLM Benchmark Inter-Phase Delay Analysis

Phase 0 (benchmark_report_results.json_0.yaml): Start = 1773719532.20, Stop = 1773720132.20 (Duration: 600.0s)
Gap between Phase 0 and 1: 3.67 seconds
Phase 1 (benchmark_report_results.json_1.yaml): Start = 1773720135.87, Stop = 1773720735.87 (Duration: 600.0s)
Gap between Phase 1 and 2: 5.33 seconds
Phase 2 (benchmark_report_results.json_2.yaml): Start = 1773720741.20, Stop = 1773721341.20 (Duration: 600.0s)
Gap between Phase 2 and 3: 9.39 seconds
Phase 3 (benchmark_report_results.json_3.yaml): Start = 1773721350.59, Stop = 1773721950.59 (Duration: 600.0s)

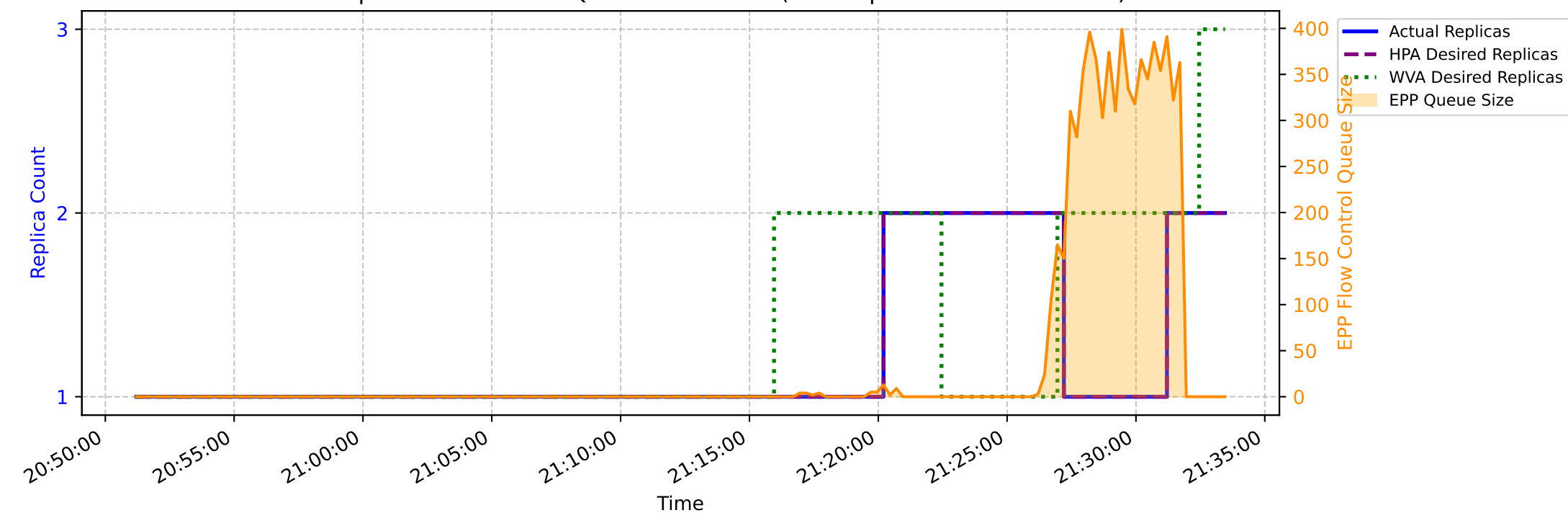
KV Cache Usage Percentage Over Time (Namespace: wva-bench-test)



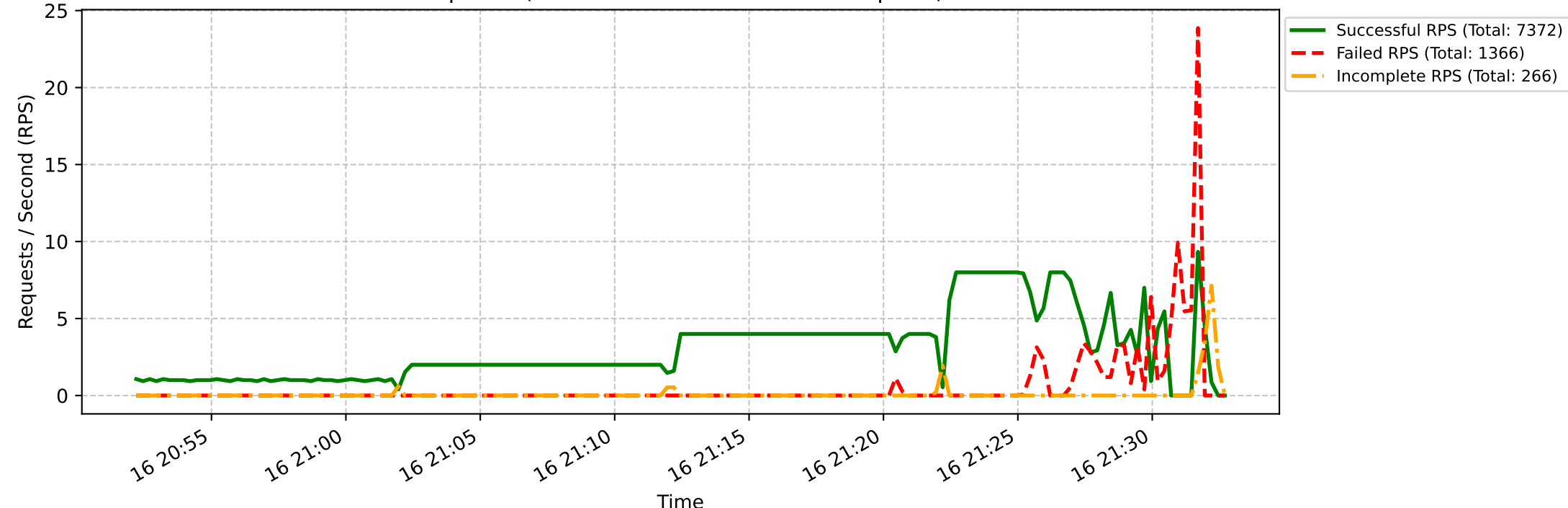
Number of Requests Waiting Over Time (Namespace: wva-bench-test)



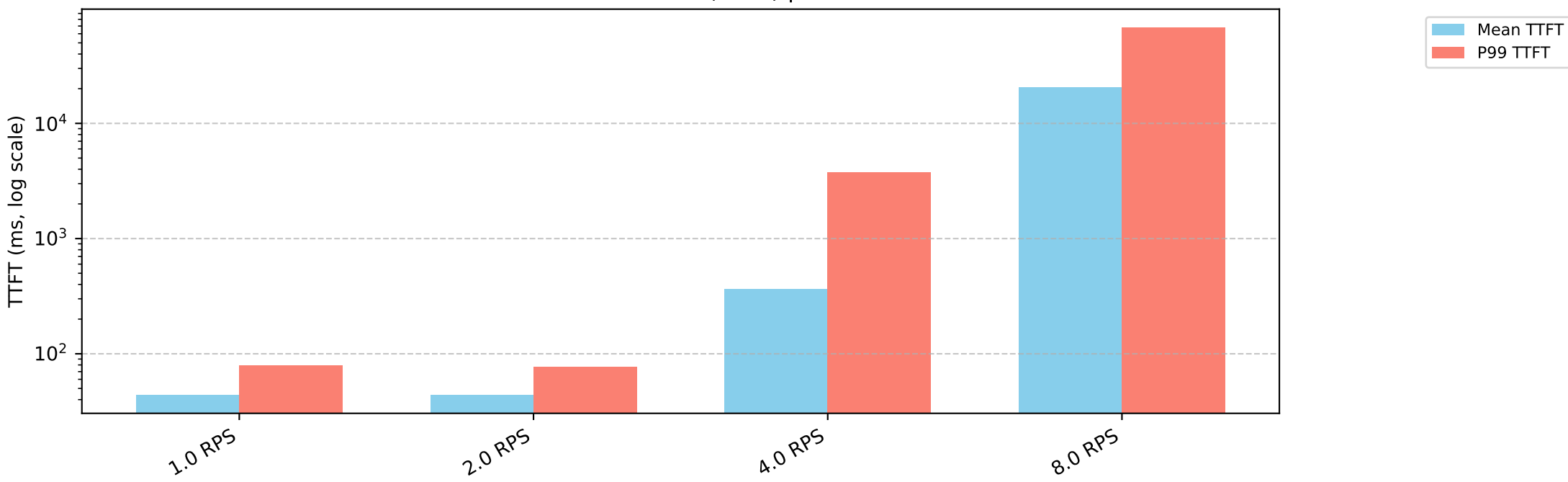
Decode Replica Count & EPP Queue Over Time (Namespace: wva-bench-test)



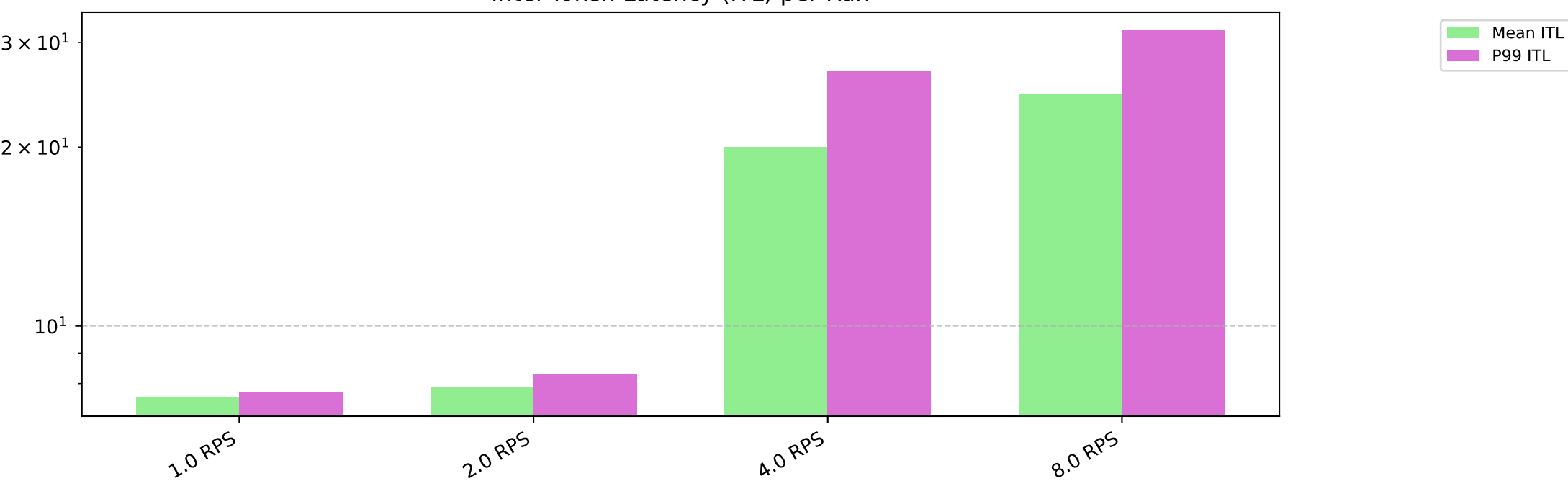
GuideLLM Requests (Succeeded vs Failed vs Incomplete) Over Time



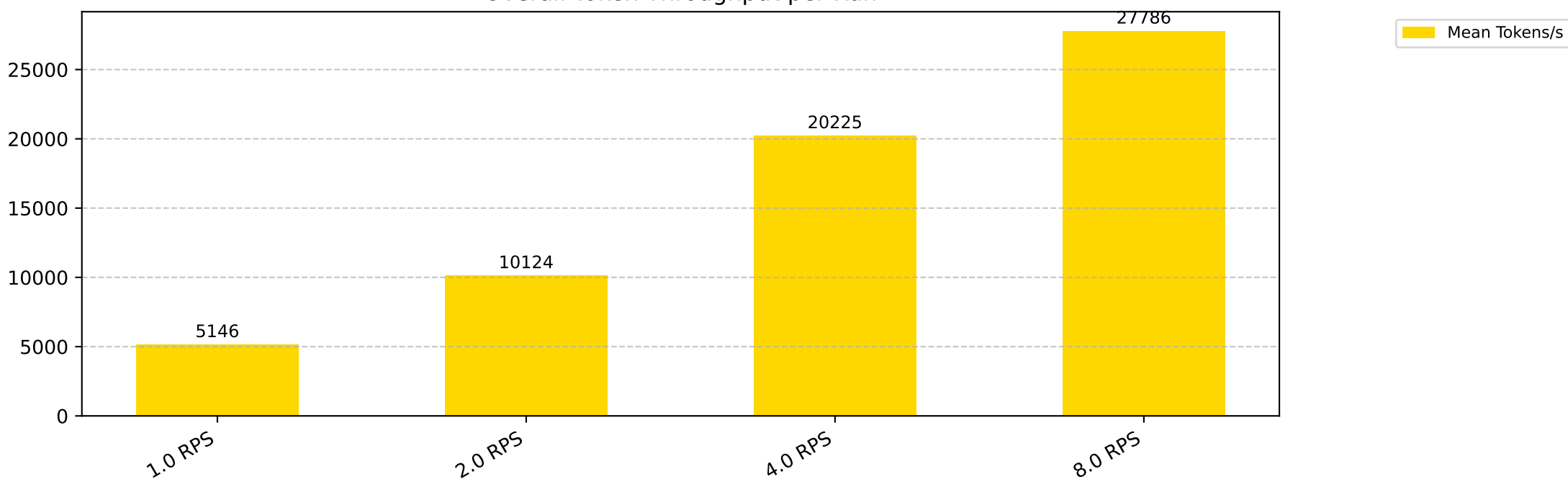
Time To First Token (TTFT) per Run



Inter-Token Latency (ITL) per Run



Overall Token Throughput per Run



Request Concurrency vs End-to-End Latency Profile

